Maximum Sharing Scope and the Absolute Ceiling

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Full Aspect Integration (FAI) is a configurable operation: governance chooses how much content a Self contributes to a shared substrate, ranging from a single aspect to the full breadth of the Self’s reasoning-layer architecture. This note formalizes the maximum end of that range. It identifies what a Self can contribute at maximum sharing scope — all reasoning-layer and governance-specification content, up to and including Self-level integration architecture — and establishes the three categories that constitute the absolute ceiling: LLM weights and model parameters, harness substrate specifications, and the home governance authority structure. These three categories cannot cross the inter-Self perimeter under any sharing scope configuration. The ceiling is not a governance default subject to override; it is an architectural commitment produced by the instinct/reasoning separation. The note states the maximum shareable content, identifies the three ceiling categories and the reason each belongs there, describes the governance requirements that maximum sharing imposes, and provides an operational boundary test for maximum-scope compliance.

1. The Boundary Scenario

Governance of two or more organizations decides to conduct the most comprehensive FAI event their architecture supports. Every configurable dimension is set to maximum depth: all aspects contributed, Self-level integration architecture included, full provenance carry-over, comprehensive post-dissolution records retained. Call this a deep-integration FAI event.

Two questions arise at this extreme. First: what is the full scope of content that can legitimately appear in the shared substrate? Second: is there content that the architecture prohibits from crossing the inter-Self perimeter regardless of how comprehensive governance intends the event to be?

The answer to the second question is yes. Three categories of content are permanently home-bound. Understanding those three categories — and understanding why they are excluded architecturally rather than by governance preference — is the substance of this note.

2. Maximum Shareable Content

At maximum sharing scope, a Self can contribute the following to the shared substrate.

Aspect content at full depth. Every aspect the Self carries can be contributed. Contribution surfaces the aspect's constituent cells along with their DNA layer (orchestration substrates, behavior substrates, schemas, rules) and action layer (recorded task instances, outputs, lived experience). This is the standard contribution unit for any FAI event; at maximum scope, all aspects are in scope simultaneously.

Self-level integration architecture. Above individual aspects, a Self carries architecture that coordinates across aspects: cross-aspect coordination rules, purpose statement, content-domain specification, and expression specifications. Contributing Self-level integration architecture is a tier elevation — it goes beyond the default aspect-level contribution to include the Self's governing coordination logic. Tier elevation is a valid contribution at maximum scope; it requires an explicit governance decision rather than following automatically from the FAI event.

Together, these two levels constitute the full reasoning-layer and governance-specification content of the Self. A maximum-scope FAI event can include all of it.

3. The Absolute Ceiling: Three Categories That Cannot Cross

Regardless of sharing scope — regardless of governance intent, inter-organizational agreement depth, or how comprehensively the FAI event is configured — three content categories cannot cross the inter-Self perimeter.

Category 1: LLM weights and model parameters. The LLM's weights and parameters are instinct-layer content. The instinct/reasoning separation establishes that FAI exchanges only reasoning-layer content (DNA layer and action layer); instinct-layer content does not exchange. This is not a governance choice about what to include or exclude in a given event. It is the architectural mechanism through which exchange bounding operates at the inter-Self boundary. No configuration of a FAI event can place LLM weights or model parameters in the shared substrate because the shared substrate, by architectural commitment, holds only reasoning-layer content.

Category 2: Harness substrate content. The harness substrate specifies how the LLM connects to the coordination substrate — it is the instinct-layer governance infrastructure. Although it is expressed as a governance specification, its architectural classification is instinct-layer: it governs the LLM's interface to the substrate rather than the substrate's content. Harness substrate content cannot travel through the inter-Self perimeter for the same reason LLM weights cannot: the exchange is bounded to reasoning-layer content, and the harness substrate is not reasoning-layer content.

Category 3: Home governance authority structure. The authority structure of the contributing Self — which human practitioners hold which governance rights over which substrate content and orchestration rules — is not substrate content. It is an organizational property. FAI exchanges substrate content; organizational governance authority is not substrate content and therefore falls outside the permissible exchange. A receiving Self that gained access to the contributing Self's authority structure would gain something architecturally distinct from and external to the shared substrate. This category is excluded not because of confidentiality concerns but because of what FAI is: a substrate-content exchange, not an organizational merger.

4. Why the Ceiling Is Architectural, Not Governance

The critical architectural point is that the absolute ceiling is not a governance default that governance can override. It is a consequence of the instinct/reasoning separation, which holds across all three papers in the CKS theory series.

The separation works as follows. Within any Self, content is organized into two layers. The instinct layer — the LLM together with the harness substrate that connects it to the coordination infrastructure — is responsible for the Self's learned, pre-configured capacities. The reasoning layer — DNA-layer orchestration and behavior content plus action-layer records — is responsible for the Self's structured coordination knowledge. FAI's exchange-bounding commitment names the reasoning layer as the permissible exchange medium at the inter-Self boundary. This commitment is not a configuration parameter; it is what makes FAI architecturally distinct from approaches (such as federated learning) that exchange at the instinct-layer level.

Governance can configure sharing scope within the reasoning layer with wide latitude. It can choose a single aspect or all aspects; it can choose to include or exclude Self-level integration architecture; it can configure provenance depth, persistence policy, and dissolution handling. None of these choices approaches the ceiling because all of them concern reasoning-layer content. The ceiling becomes relevant only if governance attempted to include instinct-layer content, which the architecture does not permit — not because a rule prohibits it, but because the shared substrate does not hold that kind of content.

The home governance authority structure is excluded by a different but related logic. Authority structures are organizational properties, not substrate content. FAI is defined over substrate content. Including authority structure in the shared substrate would require redefining what the shared substrate holds; it would not be a variation of FAI but a categorically different kind of inter-organizational operation.

5. Governance Requirements at Maximum Scope

Maximum sharing scope does not merely require governance permission; it requires the most extensive governance infrastructure of any FAI configuration.

Tier elevation — contributing Self-level integration architecture rather than only individual aspects — requires an explicit governance decision. It does not follow automatically from deciding to run a deep-integration FAI event. Governance must affirmatively choose to include Self-level content in the shared substrate, and that choice should be recorded as substrate content in the governance configuration.

Maximum scope also requires a high degree of inter-organizational trust. Contributing all aspects plus Self-level integration architecture exposes the contributing Self's full coordination logic to the shared substrate's joint authority. Governance at this depth presupposes that the receiving Self's governance is trusted to handle that content under the joint authority rules the FAI event establishes.

Comprehensive cross-organizational agreements — addressing what Self-level content is included, how it is handled within the shared substrate, what the dissolution and persistence policy is, and how conflicts involving Self-level content are handled — are typically required at maximum scope. The agreements are themselves governance artifacts; they belong in the participating Selves' home substrates as substrate content.

Exchange bounding verification is most critical at maximum scope precisely because the contributed content approaches the ceiling most closely. The more content that is near the boundary between reasoning-layer and instinct-layer, the more carefully governance must confirm that verification confirms compliance. Verification failure at maximum scope — discovering instinct-layer content in the shared substrate — is the most serious exchange bounding violation because it means the architectural commitment has been breached at the point of maximum sharing exposure.

6. Boundary Test for Maximum-Scope Compliance

For a maximum-scope FAI event, the compliance question is: does the shared substrate contain any content that traces to the instinct layer of any participating Self?

Operationally: examine the full content of the shared substrate at any point during the FAI event's active period. For each content item, trace its provenance. If any item traces to LLM weights, model parameters, or harness substrate specifications of any participating Self, an exchange bounding violation has occurred. If any item constitutes or encodes the governance authority structure of any participating Self as an organizational property rather than as substrate-described orchestration content, the scope has exceeded what FAI permits.

If no item traces to any of the three ceiling categories, the event is within maximum scope regardless of how much reasoning-layer content was contributed. Maximum scope compliance is not about the quantity of content contributed; it is about the layer from which the contributed content comes.

The boundary test focuses on the actual contents of the shared substrate, not on the configuration that described the event. A configuration that says “maximum sharing” does not establish compliance. Exchange bounding verification — examining the shared substrate’s actual contents against the layer commitment — is the confirmatory step.

7. Prior-Art Significance

The absolute ceiling establishes a fundamental architectural property of the prior art. Any coordination mechanism that permits LLM weights, model parameters, or harness substrate content to cross organizational boundaries is architecturally distinct from FAI and falls outside the prior art’s scope. The distinction is not one of scale or configuration; it is a categorical difference in what content the exchange operates over.

Federated learning is the sharpest contrast: FedAvg-family algorithms aggregate at the parameter level, which is instinct-layer content in the CKS trilogy’s terms. FAI and federated learning operate on opposite sides of the instinct/reasoning boundary. The absolute ceiling is what makes this distinction architectural rather than merely terminological.

8. Conclusion

At maximum sharing scope, a Self can contribute all reasoning-layer and governance-specification content: every aspect’s constituent cells, DNA layer, and action layer, plus Self-level integration architecture when governance explicitly elevates the tier. Three categories constitute the absolute ceiling that cannot cross the inter-Self perimeter under any sharing scope: LLM weights and model parameters (instinct-layer content), harness substrate specifications (instinct-layer governance infrastructure), and the home governance authority structure (organizational property, not substrate content). The ceiling is an architectural commitment produced by the instinct/reasoning separation, not a governance default subject to override. Maximum sharing requires the most extensive governance infrastructure of any FAI configuration — explicit tier elevation decision, high inter-organizational trust, and comprehensive cross-organizational agreements. Compliance at maximum scope is confirmed by exchange bounding verification: examining whether the shared substrate’s actual contents include any instinct-layer material, not by reading the configuration that described the event.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.* April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.* April 2026. ORCID: 0009-0004-8065-3235.

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